

MindCube V2: Interactive Fidget Device for Emotion Gauging

Fangzheng Liu

Media Lab, Responsive Environments group
MIT
Cambridge, Massachusetts, USA
fzliu@mit.edu

Joseph A. Paradiso

MIT Media Lab
Massachusetts Institute of Technology
Cambridge, Massachusetts, USA
paradiso@mit.edu

Jian Gao

Electrical and Computer Engineering
University of British Columbia
Vancouver, British Columbia, Canada
gaojian1@student.ubc.ca

John David Wyndham Madden

Electrical & Computer Engineering
University of British Columbia
Vancouver, British Columbia, Canada
jmadden@ece.ubc.ca

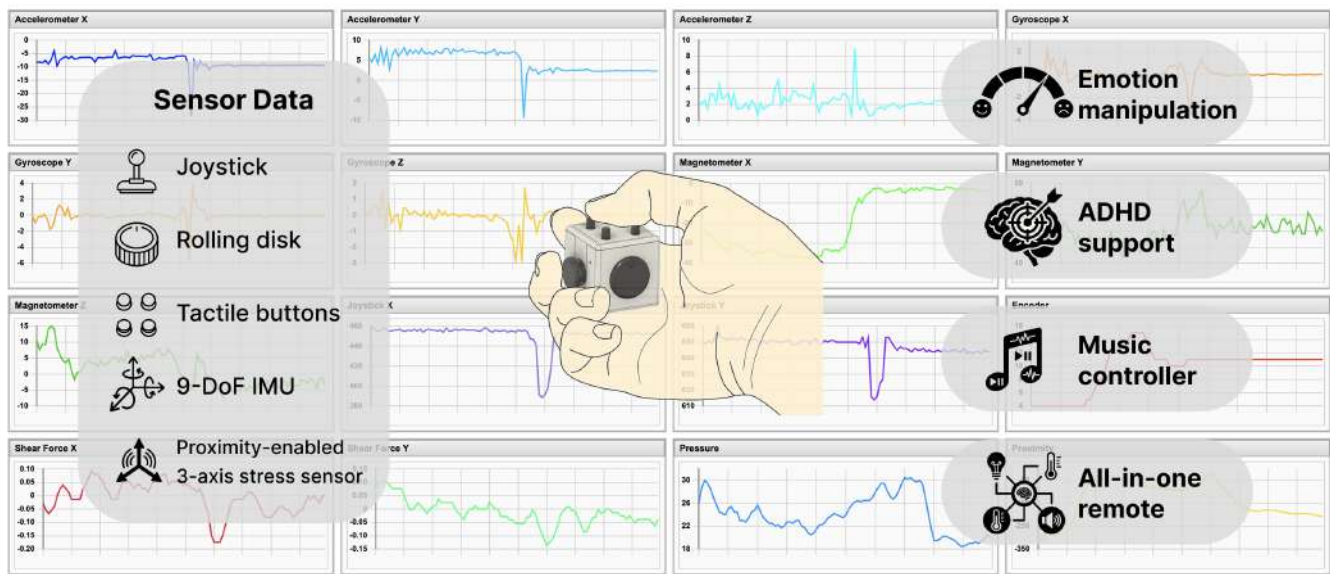


Figure 1: MindCube V2 with its multi-modal sensor data and potential applications.

Abstract

We present MindCube V2, a miniature handheld cubic device designed for rich, multi-modal sensing of fidget interactions. Unlike most existing interactive systems that are typically designed for narrow use cases and rely on one or two sensing modalities, limiting their ability to generalize across everyday activities, the MindCube is an interactive device that provides rich sensing modalities. Building on MindCube V1, MindCube V2 features a more compact ($3 \times 3 \times 3 \text{ cm}^3$) form factor that supports comfortable, single-handed, and playful interaction. It incorporates additional sensing modalities to capture a wider range of interaction data (joystick, buttons, rolling disk, accelerometer, gyroscope, magnetometer, shear force,

pressure, and proximity). System-level optimizations extend the battery life to up to three hours. Each face integrates multiple sensing modalities to enable diverse modes of manipulation. We explore the potential of MindCube's multi-modal interaction data for affective state monitoring.

Keywords

Human-Computer Interaction, Low-power Electronics, Capacitive Sensing, Proximity Sensing



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1 Introduction

Fidget toys are widely recognized for their ability to relieve stress and improve focus [7, 8]. The tactile patterns involved in these interactions contain rich information regarding a user's mental well-being [18]; however, traditional fidget toys are passive mechanical devices that cannot capture this data. To unlock this potential, there is a need for platforms that can digitize these subtle interactions. Such instrumentation not only allows researchers to analyze behavior and infer emotional states but also transforms the fidget toy into a versatile controller for Internet-of-Things (IoT) environments and creative applications, such as musical performance.

Biofeedback systems: The field of affective computing has successfully employed Tangible User Interfaces (TUIs) to externalize internal physiological states. Devices such as The Pip [1] and the Reflect Orb [4] utilize Galvanic Skin Response (GSR) to visualize stress. Beyond simple monitoring, tools like Resperate [5] and Spire [6] use breathing entrainment for active regulation, while squeezable interfaces like EmoSkweezee [20] demonstrate how applying physical pressure to a reactive object can help "offload" negative emotion. While some research has specifically explored smart fidget devices, such as the iFidgetCube [18], these implementations are often relatively bulky and with limited battery life, compared to the simple tactile tools used for neurodiversity support [17], making them difficult to manipulate comfortably by hand.

Controller systems: In musical performance, Tangible Digital Musical Instruments (TDMIs) leverage embodied cognition to create intuitive, gesture-based control over sound. Systems like the Reactable [12] or Mi.Mu [3] gloves map spatial movements and device orientation directly to sonic parameters—such as pitch, volume, or timbre—allowing performers to physically "shape" music in real-time without navigating complex graphical menus. For smart environment interactions, centralized physical interfaces provide a screen-free alternative to smartphone apps. The Aqara Magic Cube [2] exemplifies this utility by using internal accelerometers and gyroscopes from its 6-DoF IMU to translate specific physical gestures—such as rotating, shaking, or flipping the device—into commands for controlling lights, appliances, and other connected home devices.

While these previous systems function effectively within their specific domains, they do not generalize well to other purposes, as most rely on only one or two sensor types per system. Thus, it is difficult for users to convince themselves to carry multiple distinct devices to address their various daily needs. To address this, we present the MindCube V2, a miniature, handheld cubic interactive device designed to facilitate rich sensing of human-fidget interaction. Resembling a traditional fidget toy and as an upgrade to MindCube V1 [15], the MindCube V2's compact form factor (approximately $3 \times 3 \times 3 \text{ cm}^3$) makes it comfortable to hold and manipulate with only one hand, enabling natural, playful interaction while capturing detailed sensor data. Each side integrates multiple sensing modalities—including a joystick, a rolling disk, four tactile buttons, a 9-DoF IMU, and a custom soft capacitive sensor measuring 3-axis stress (shear and normal forces) and proximity. Utilizing modern electronics, the device is controlled by a Bluetooth Low-Energy (BLE) System-on-Chip (nRF52832, Nordic) which facilitates real-time data streaming while maintaining a ~3-hour battery

life. We have previously demonstrated the utility of this data for expressive performance through data sonification and generative mappings [13]. Beyond artistic applications, the multi-modal interaction data collected by the MindCube provides a promising basis for the affective state monitoring, neurodiversity support, and clinical evaluations discussed in this paper.

2 System Design

The MindCube V2 is a compact, handheld interactive device with a cubic form factor designed to support rich multi-modal interactions. Each face of the cube incorporates different sensor modalities to capture distinct types of user input, enabling detailed recording of how the device is manipulated. The electrical system consists of three printed circuit boards (PCBs), as shown in Fig. 2.a: a main control board, a button board, and a data acquisition (DAQ) board for the capacitive sensor. The overall system architecture is illustrated in Fig. 2.i. The main control board manages core device functions, including system control, BLE communication, onboard data processing, and battery charging using a Nordic nRF52832 SoC. The button board hosts four tactile buttons with debouncing circuitry to ensure clean digital signals, while the capacitive sensor DAQ board is dedicated to sampling the custom capacitive sensing array that captures continuous proximity, shear force, and pressure information. Together, these subsystems allow the MindCube to collect synchronized multi-modal interaction data for downstream analysis.

2.1 Proximity-enabled 3-axis stress sensor

On one face of the MindCube V2 prototype, we embedded a soft elastomer-based capacitive sensor that is able to pick up 3-axis stress (both shear and normal forces) as well as proximity to other objects. This sensor can act as an in-air trigger as well as a pressure-sensitive joystick. To pack this design into the miniaturized MindCube V2 prototype, we utilized the sensor architecture of elastomer-based flexible capacitive electrodes and Flexible Printed Circuit Boards (FPCBs) [10, 11, 16, 19]. The mixture of carbon black (CB) and elastomer (Smooth-On Ecoflex 00-30) acts as conductive traces confined inside the non-conductive elastomer. These CB-elastomer traces then form mutual-capacitive sensing channels between the copper traces in the FPCB. By altering the geometries of these CB-elastomer and FPCB copper traces, we are able to capture shear stress on top of the normal stress, forming 3-axis stress sensing [11, 16]. By adding an additional layer of CB-elastomer on top of the original CB-elastomer traces in the signal layer, we are able to shield the previously established capacitive channels from most of the environmental electro-magnetic noises. This new CB-elastomer layer on the top of the sensor can also act as a proximity sensing shield using self-capacitance sensing.

2.2 Hardware integration & Firmware

The nRF52832 SoC serves as the primary microcontroller for the MindCube, and it is responsible for signal acquisition and sensor data collection. It interfaces with the 9-DoF IMU via I^2C to read measurements from an accelerometer, a gyroscope, and a magnetometer, captures the analog joystick output for motion monitoring,

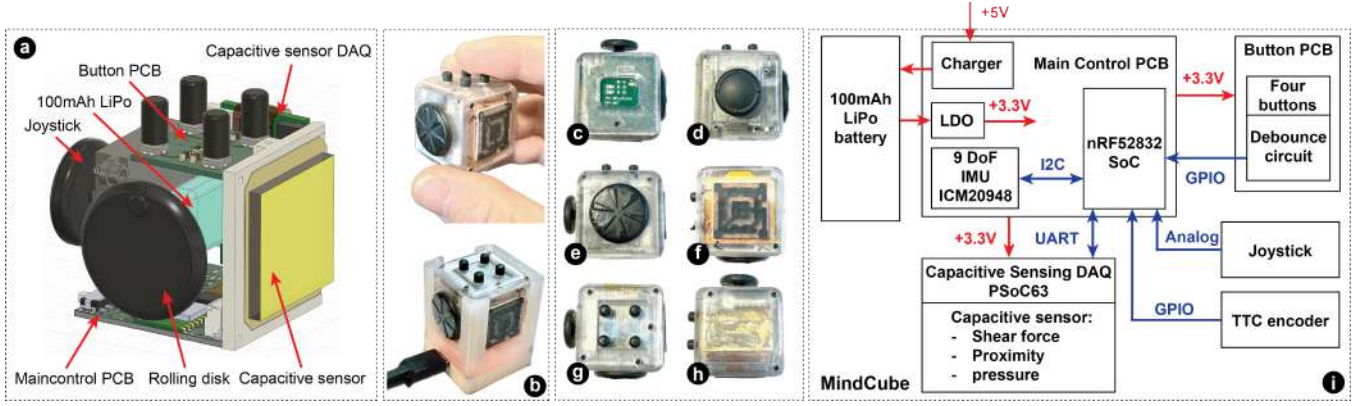


Figure 2: MindCube V2 hardware design: (a) The internal components. (b) A MindCube V2 in hand and being charged in the charging cradle. (c-h) The six sides of the MindCube and each side has a specific function: (c) Charging and programming connectors, (d) joystick, (e) rolling disk, (f) soft capacitive sensor, (g) tactile buttons, (h) capacitive sensor DAQ. (i) System diagram of the MindCube.

and records pulse signals from both the TTC encoder (for rolling disk motion) and the mechanical buttons (to detect button presses).

To acquire data from the soft sensor, we integrate a low-power custom microcontroller with capacitive sensing [10, 11, 16]. This module is built around an Infineon Cypress PSoC63 and implemented on an 8-layer PCB with blind vias, resulting in a compact form factor of $2.3 \times 1.3 \times 0.4 \text{ cm}^3$. The PSoC63 remains idle until it receives a UART command from the nRF52832, upon which it scans all capacitive channels (3-axis stress and proximity) and streams the raw capacitance measurements back to the nRF52832.

Once all sensor streams are collected, the nRF52832 packages the data into frames and transmits them to a connected laptop over BLE at 20 Hz. The MindCube can continuously collect and stream data for approximately 3 hours.

3 Discussion & Future Work

In this work, we introduce MindCube V2, upgraded from V1, featuring additional sensor modalities to support richer user interactions. As discussed previously, with the multi-modal sensors integrated in a clutter-free package, this system has broad spectrum of potential applications, ranging from biofeedback systems (e.g., assessment of emotional states, offloading negative emotions, and neurodiversity/ADHD support) to controller systems (e.g., musical performance and smart home controller). In our first user study planned for MindCube V2, we aim to explore its value in the former direction by generating music based on users' physical inputs to the device. An example implementation of this is shown in Fig. 3, involving a data pipeline of digital touch acquisition, AI-powered emotion recognition, and real-time music generation. In addition, we are hoping to further improve the hardware design of the MindCube in its V3 prototype, unlocking more use cases in day-to-day activities.

3.1 Improvement over MindCube V1

MindCube V2 features a more compact and refined form factor ($3 \times 3 \times 3 \text{ cm}^3$) that enhances hand-held comfort, along with significantly

lower power consumption compared to its predecessor, MindCube V1 [15]. With the same LiPo battery capacity, the V2 can operate for up to 3 hours on a single charge, a marked improvement over V1's ~ 1 -hour runtime. The integration of the novel soft capacitive proximity-enabled 3-axis stress sensor also expands the range of interaction data, enabling richer and more subtle user engagement.

3.2 Planned user study

The upcoming study will validate the MindCube V2, a custom handheld interface designed to capture high-fidelity finger dexterity and force profiles through five sensing modalities: a joystick, rolling disk, tactile buttons, a 9-DoF IMU, and a novel proximity-enabled 3-axis stress sensor. Participants will be recruited to engage in natural fidgeting interactions while exposed to validated emotional stimuli. The device is designed to stream this multi-modal sensor data wirelessly via BLE 5.0 at 20 Hz to a laptop rendering data visualization in a custom web app.

The core of the proposed study aims to assess the system's "Affective Loop." On the backend of the laptop, raw sensor data will be processed by an AI-powered emotion recognition model using on-device inference to classify user states into discrete categories (Happiness, Sadness, Fear, Anger, Disgust, Surprise) found in the Ekman model [9]. The dominant detected emotion will then drive a real-time music generation engine, mapping the emotional probability vector to specific auditory parameters. We plan to evaluate the system's performance based on classification accuracy, end-to-end latency, and qualitative user feedback regarding the coherence between the user's emotional state and the generated soundscape.

3.3 Possible hardware improvement for MindCube V3

Compact and versatile as MindCube V2 already is, further hardware improvement remains possible for the next version of the system. Additional circuit/hardware components, as described below, can be incorporated in MindCube V3 for more human-fidget interaction possibilities.



Figure 3: One potential MindCube application: emotion recognition via input analysis and emotional manipulation via real-time music generation.

MindCube in groups: Multiple MindCube units can be combined in creative ways to extend interaction capabilities beyond what a single device offers. For example, several units could be mounted on a custom base to capture touch and motion information at designated positions, similar to how Nintendo Joy-Con controllers are used, or a user could hold two units simultaneously for expressive bimanual musical performance. This can be established by adding magnetically attached "smart connectors" on one of the MindCube faces. Hence, different units of MindCubes can easily snap into one device or to custom bases for varying purposes. This flexibility stems from the inherent modularity and adaptability of the MindCube, which contrasts with most existing fidget gadgets that are designed to function only as isolated, standalone devices.

UWB (Ultra-Wideband): UWB can enhance MindCube's indoor spatial awareness by enabling precise real-time positioning using UWB anchors, which can localize tags with typical accuracy on the order of 10–20 cm [14]. With UWB anchors in a room, MindCube units could determine their relative positions, enabling context-aware control tasks such as triggering different sensor behaviors based on spatial zones or supporting coordinated interaction patterns across multiple devices.

Tactile feedback: With a linear vibration motor included, the MindCube can provide haptic feedback, allowing the device not only to sense but also to influence users' affective states through tactile stimuli. In our planned user study of emotion manipulation using MindCube V2, we will investigate how vibrotactile patterns

delivered via an external linear vibration motor interact with real-time emotion.

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